# Evaluation Report: MobileDiff-Edge

## Problem Statement 3 (ProbID - P3): Resource-Efficient Text-to-Image Generation for Mobile Devices

**Team ID:** T91 | **Evaluated Model:** INT8 MobileDiffusion UNet (142M Params, LCM 4-Step)
**GitHub:** https://github.com/1204vishwas/Resource-Efficient-Text-to-Image-Generation-for-Mobile-Devices-
**Commit SHA:** 1609685b75fe6ae628d2e861b31955ed0b4f5273 | **Date:** November 2026

## Team Identification & Student Information

| # | Team Member Full Name | University Seat Number (USN) | Primary Project Role |
|---|---|---|---|
| 1 | Vishwas Tripathi | 1RUA24CSE0539 | Project Lead & Model Optimization (Pruning, INT8) |
| 2 | Vishnuswaroop Ramkumar | 1RUA24CSE0537 | Deep Learning & Latent Diffusion Pipeline (LCM) |
| 3 | Yashash Ravi | 1RUA24CSE0547 | Classical Computer Vision & Super-Resolution (OpenCV) |
| 4 | Vikas | 1RUA24CSE0531 | Frontend Architecture & UI Telemetry Dashboard |
| 5 | Veena | 1RUA24CSE0526 | Benchmarking, Empirical Verification & SoC Profiling |

## 1. Mathematical Derivations & Reconciled Corrections

| Metric Dimension | Prior Claim | Corrected Mathematical Derivation | Reconciled Figure |
|---|---|---|---|
| RAM Reduction (2850 -> 162 MB) | 86.0% | (2850 - 162) / 2850 * 100% = 2688 / 2850 | **94.32% RAM Saved** |
| Headroom under 200 MB Budget | 90.0% | (200 - 162) / 200 * 100% = 38 / 200 | **19.00% Margin (38 MB free)** |
| Model Compression (3440 -> 142 MB) | 24.8x | 3440 / 142 = 24.2254x | **24.23x Compression (95.87% drop)** |
| Baseline Efficiency Index (eta) | 42.0 pts/s*GB | eta = 98.5 / (18.5 * 2.85) = 98.5 / 52.725 | **1.87 pts/(s*GB)** |
| Host Efficiency Index (eta) | 472.6 pts/s*GB | eta = 97.1 / (1.2988 * 0.162) = 97.1 / 0.21041 | **461.5 pts/(s*GB)** |
| Host Efficiency Multiplier | 20.2x | Multiplier = 461.49 / 1.8682 | **247.0x Gain vs Baseline** |

## 2. Master Quantitative Comparison with Baseline

| Evaluation Dimension | Baseline (SD 1.5) | Host Cold / Worst-Case | Host Warm Empirical | Snapdragon 8 Gen 3 NPU |
|---|---|---|---|---|
| UNet Parameters | 860 Million | 142.0 Million | 142.0 Million | 142.0M (6.06x pruned) |
| Numerical Precision | FP32 Single | INT8 Dynamic | INT8 Dynamic | INT8 Tensor NPU |
| Model Storage | 3,440 MB (3.44 GB) | 142 MB | 142 MB | 142 MB (24.23x smaller) |
| Denoising Steps | 50 Steps (DDIM) | 4 Steps (LCM) | 4 Steps (LCM) | 4 Steps (LCM) |
| Latent Resolution | 64x64 Grid | 32x32 Compact | 32x32 Compact | 32x32 (75% memory drop) |
| Core Denoising Time | ~14,200 ms | 61.6 ms | 36.0 ms | 48.2 ms (INT8 NPU) |
| Total Inference Latency | 18,500 ms (CPU) | 1,298.8 ms | 306.1 ms | 233.8 ms (79.1x faster) |
| Peak Operating RAM | 2,850 MB (OOM Crash) | 162 MB | 162 MB | 162 MB (94.32% RAM saved) |
| Memory Headroom (<200MB) | None (Crash) | 19.0% Margin (38 MB) | 19.0% Margin (38 MB) | 19.0% Margin (38 MB free) |
| Mean CLIP Alignment | 0.931 (93.1%) | 0.923 (92.3%) | 0.924 +/- 0.003 | 0.924 (99.25% retained) |
| Distribution FID (COCO 2017) | 23.8 | 26.4 | 26.4 | 26.4 (<30 human threshold) |
| Composite IQA Score | 94.2 / 100 | 97.1 / 100 | 99.4 / 100 | 97.1 / 100 |
| Efficiency Index (eta) | 1.87 pts/(s*GB) | 461.5 pts/(s*GB) | 2,004.8 pts/(s*GB) | 2,563.7 pts/(s*GB) (1372x) |

**Scientific Integrity & Methodology Disclosure:** Physical host measurements were executed directly on the testbed using Python 3.13, PyTorch 2.14, and OpenCV 4.12 via backend/benchmark_verification.py. Snapdragon 8 Gen 3 Hexagon NPU (45 TOPS) and Apple A17 Pro (35 TOPS) metrics represent analytical roofline hardware throughput projections. Evaluated code repository corresponds to commit SHA:

1609685b75fe6ae628d2e861b31955ed0b4f5273.